

VARIOUS WIRELESS MODULES

Module	Wireless technology support
ESP 32	Wi-Fi and BLE
ESP 8266	Wi-Fi
CY8CKIT-049	Wi-Fi-ready development board
EMW 3165	Wi-Fi
nRF8001	BLE
BM71	BLE
Bluefruit EZ-Link	BLE
RFM98W	LoRa
SX1276	LoRa
LE51-868 S	SigFox

power of system components and end-user behaviour. Transferred data rate is often significantly lower than the maximum achievable throughput, when various protocol overheads are considered.

Kamble explains the significance of throughput as the need of

low data but high coverage in Internet of Things (IoT) and M2M communications. "LoRa and SigFox technologies address longer battery life and large coverage areas for lesser throughput solutions," says Kamble.

What is next in wireless

Today's wireless systems use non-chaotic wireless signals, ranging over satellite communications, GPS navigation, mobile phones and Wi-Fi devices. These have a well-defined period and frequency, which causes physical impediments with wireless signals in open spaces. A study done a couple of years ago proposed chaotic signals for overcoming some of the constraints faced by wireless devices to improve performance. An added advantage was that these allowed designing of low-power, low-cost and small-area electronic circuits.

Chaotic signals are strongly modified by wireless physical media through which it propagates, but information transmitted by the signal is not modified. Chaotic signals are also unaffected by multipath effects, which is a major deterrent with most wireless transmissions.

Controlling chaos. Being aperiodic, irregular, having broadband spectrum, easy to generate and difficult to predict over time makes chaotic signals a very lucrative solution in sonar and radar applications. A research at Xian University of Technology in China and University of Aberdeen in the UK has demonstrated how chaotic signals could create a reliable and efficient wireless communication system.

However, initial conditions need to be ideal. The tiniest of mistakes could lead to circuits not encoding the information for transmission. With successful implementation of chaos theory, these can not only be beneficial with secure communication but also encode a binary source of information in an energy-efficient way. So maybe, we finally have a place where chaos is not all that bad. **EPY**

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THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT

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